

Aziz Fouché, Ph.D.

AI Research Scientist

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Experience

AI Research Scientist for Computational Biology

May 2024 – current

Scientia Lab

Paris, France

- Development of multimodal foundation models of biological data (tabular and image).
- Researching novel architectures, pre-training targets, and aligned benchmark tasks.
- Development of training and benchmark pipelines (PyTorch, Lightning, Metaflow...)
- Work with production-grade environments (AWS/GCP/SLURM, MLFlow, Databricks...)
- Scientific project organization, intern mentorship, hiring, communication.

Ph.D. in AI for Computational Biology

Sept. 2020 – Nov. 2023

Institut Curie (U900), supervised by A. Zinovyev and O. Delattre

Paris, France

- Developing ML applications handling high-dimensional single-cell transcriptomics datasets.
- Methods for automatic batch effect correction and signal deconvolution for tabular data.
- Jury: G. Peyré, V. Boeva, D. Surdez, A. Zinovyev, O. Delattre, J.-P. Vert.

Teaching Assistant – Algorithms and Java Programming

Sept. 2020 – June 2021

Université de Paris

Paris, France

Research Internship in Epidemiology

Feb. – July 2020

Institut Pasteur (UMR 2000), supervised by S. Cauchemez and M. Lecuit

Paris, France

- Modeling the infection dynamics of *Listeria monocytogenes* in mice using Bayesian inference.

Research Internship in Single-Cell RNA-seq Analysis

May – July 2019

Institut Curie (U900), supervised by A. Zinovyev

Paris, France

- Analysis of transcriptional dynamics at the single-cell level of Ewing sarcoma tumors; studying tumoral heterogeneity sources.

Research Internship in Genetic Diseases

Feb. – June 2018

ULB/VUB, supervised by T. Lenaerts and G. Smits

Brussels, Belgium

- Development and explainability of random forests to predict the effect of gene-gene interaction in oligogenic diseases.

Research Internship in Distributed Algorithmics

May – July 2016

LABRI, supervised by A. Zemmari

Bordeaux, France

- Development of distributed algorithms using a communication protocol of size $O(1)$.

Education

Ph.D. in Computational Biology – Sorbonne Université / Institut Curie

2020–2023

Licence in Biology – ENS Paris-Saclay

2018–2019

École Normale Supérieure Paris-Saclay – Computer Science

2017–2020

Licence in Mathematics & Computer Science – Université de Bordeaux

2014–2017

- [1] A. Fouché, Y. Cattan, C. Claye, M. Corney, P. Marschall, K. El Kanbi, J. Duquesne, “EVA-RNA: A Scaling Cross-Species Transcriptomic Foundation Models for Immunology & Inflammation” *Learning Meaningful Representations of Life, ICLR Workshop*, 2026.
- [2] K. El Kanbi, Y. Cattan, P. Marschall, M. Corney, A. Fouché, V. Bouget, J. Duquesne, “A Comprehensive Benchmark for Transcriptomic Foundation Models in Immunology and Inflammatory Drug Development” *Learning Meaningful Representations of Life, ICLR Workshop*, 2026.
- [3] E. Bandasack, ..., A. Fouché, P. Marschall, “EVA: Towards a universal model of the immune system” *arXiv*, 2026.
- [4] A. Fouché, M. Corney, P. Marschall, F. Strozzi, B. Verstockt, T. Raine, et al. “Facing the valley of death in IBD: a ML approach to predict clinical efficacy of drug candidates from preclinical data” *J. Crohn’s and Colitis* 20 (Suppl. 1), 2026.
- [5] A. Fouché, A. Bruley, M. Corney, P. Marschall, V. Bouget, J. Duquesne. “Closing the gap between the biology and the clinic with a foundation model of immunology and inflammation” *Foundation Models for Life Sciences, ICML Workshop*, 2025.
- [6] A. Fouché, A. Bruley, M. Corney, P. Marschall, V. Bouget, J. Duquesne. “EVA: a Foundation Model Advancing Translational Drug Development in Immuno-Inflammation” *bioRxiv*, 2025.
- [7] A. Fouché, A. Zinovyev. “Omics data integration in computational biology viewed through the prism of machine learning paradigms” *Frontiers in Bioinformatics* 3, 1191961, 2023.
- [8] A. Fouché, L. Chadoutaud, O. Delattre, A. Zinovyev. “Transmorph: a unifying computational framework for modular single-cell RNA-seq data integration” *NAR Genomics and Bioinformatics* 5 (3), lqad069, 2023.
- [9] E.M. Mirkes, J. Bac, A. Fouché, S.V. Stasenko, A. Zinovyev, A.N. Gorban. “Domain adaptation PCA: base linear method for learning with out-of-distribution data” *Entropy* 25 (1), 33, 2022.
- [10] A. Zinovyev, M. Sadvovsky, L. Calzone, A. Fouché, C.S. Groeneveld, et al. “Modeling progression of single cell populations through the cell cycle as a sequence of switches” *Frontiers in Molecular Biosciences* 8, 793912, 2022.
- [11] A. Fouché, A. Zinovyev. “Unsupervised weights selection for optimal transport based dataset integration” *bioRxiv*, 2021.
- [12] M.M. Aynaoud, O. Mirabeau, N. Gruel, et al. “Transcriptional programs define intratumoral heterogeneity of Ewing sarcoma at single-cell resolution” *Cell Reports* 30 (6), 1767–1779, 2020.
- [13] N. Versbraegen, A. Fouché, et al. “Using game theory and decision decomposition to effectively discern and characterise bi-locus diseases” *Artif. Intell. Med.* 99, 101690, 2019.
- [14] A. Fouché, N. Versbraegen, et al. “Unravelling the spectrum of bi-locus diseases” *Joint ICML and IJCAI 2018, Workshop of Computational Biology*, 2018.